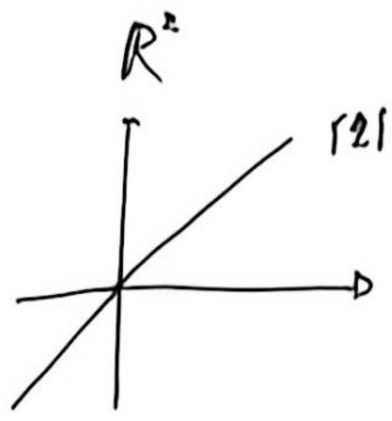
Midterm 2042 复习讲座.

1. subspace $W \subseteq V$

i) W 是子集

ii) W 是向量空间 → 封闭



过原点O的线 一个R^2的空间

$y_1 = kx_1$, $y_1 = kx_1$

$y_2 = kx_2$ $y_2 = \alpha kx_1 = k(\alpha x_1)$

$y_3 = y_1 + y_2 = k(x_1 + x_2)$

2. basis 替换定理 dimension (finite), 基的向量数量永远相等. proof: 替换: $Span(G) = V$.

↓ n个向量

取一组线性无关向量 L, m个

$m \le n$

从G里去掉 m个, 换成L,

换完这组依然V的张成集.

设 β_1, β_2 均为V基:

1° β_1 张成集, β_2 无关组

$\Rightarrow |\beta_2| \le |\beta_1|$

2° Conversely

$\Rightarrow |\beta_1| \le |\beta_2|$

so $|\beta_1| = |\beta_2|$

3. V
- T (linear transformation)
- A (matrix)

4. direct sum of W_1 & W_2

$W_1 \oplus W_2 = V \Leftrightarrow W_1 + W_2 = V,\ W_1 \cap W_2 = \{0\}$

$W_1 + W_2 = \{w_1 + w_2 \mid w_1 \in W_1, w_2 \in W_2\}$ 从W_1多一个, W_2多1个, 相加得到的会仍向量.

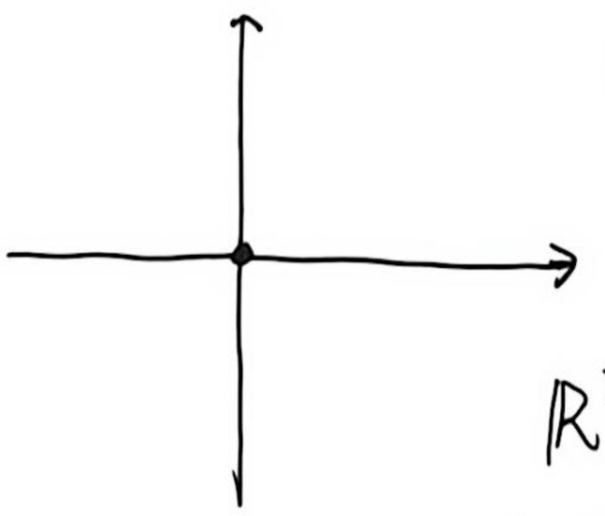
proof: $w = w_1 + w_2 = w_1' + w_2'$, $w \in W_1 \oplus W_2$

$w_1 - w_1' = w_2 - w_2' = 0$

↓ $\in W_1$ ↓ $\in W_2$

Thus $W_1 \oplus W_2$ 中的向量可被 w_1, w_2 唯一表示.

ex. 不理解就转换成坐标系. coordinates.



x轴, y轴各2个 subspaces, intersection为 {0}

↓ ↓

(a,0) (0,b)

V V

$R^2 = U + V$

$(a,b) = (a,0) + (0,b)$.

$U \cap V = \{0\} \rightarrow (a,0) = (0,b),\ a = 0 = b$.

ex. $S = \{v_1, v_2 \cdots v_k\}$

$span(S) = \{\sum a_i v_i\}$

5. span (S) = V

↓ 向量

S里所有的全部线性组合.

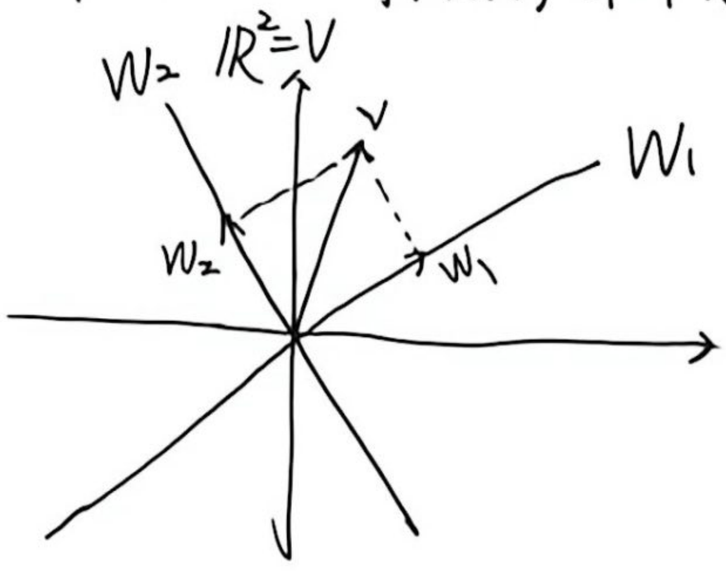
1. $\ker(T)$, $\mathrm{Im}(T)$. $\ker(T) := \{v \mid T(v)=0\}$, $\mathrm{Im}(T) := \{T(v) \mid v \in V\}$
   ↑dim ↑dim
2. nullity, rank $\qquad T: V \to W$

3. one-to-one, onto

Definite dimension vector space: $\dim V = \dim W$, $T: V \to W$ is linear.

T one-to-one $\Leftrightarrow$ onto $\Leftrightarrow$ isomorphism (proof. $\text{nullity}(T) + \text{rank}(T) = \dim V$)

4. projection 找投影作平行.



5. $T: V \to V$, $T^2 = T$ (做一次T再做一次T result=只做一次T)

prove: $V = R(T) \oplus N(T)$ i.e. $\forall v \in V$ can be written uniquely as $v = w_1 + w_2$, $w_1 \in R(T)$, $w_2 \in N(T)$

$\forall v \in V$, $v = T(v) + (v - T(v))$

$w_1 = T(v) \in \mathrm{Im}(T)$

$w_2 = v - T(v)$

$T(w_2) = T(v - T(v)) = T(v) - T^2(v) = 0$., $w_2 \in \ker(T)$

s.t. $\mathrm{Im}(T) + \ker(T) = V$

take $x \in \ker(T) \cap \mathrm{Im}(T)$.

$x = T(u)$, $T(x) = 0$., $T^2(u) = T(u) = 0$, $x = 0$.

s.t. $\ker(T) \cap \mathrm{Im}(T) = \{0\}$

Hence direct sum. $V = R(T) \oplus N(T)$

6. $W \subseteq V$

$T: V \to V$ linear

$\forall w \in W$, $T(w) \in W$

$T|_W : W \to W$. 对 $w \in W$, $(T|_W)(w) = T(w)$

↑ T在W上的限制. $\qquad T|_W : W \to W$

$T|_W : W \to V$ 永远E. But $T|_W : W \to W$ 只有当W是不变子空间时才E.

1. $\text{nullity}(T)+\text{rank}(T)=\dim(V)$

$\overset{\dim}{=}\dim \text{Ker}(T)+\dim \text{Im}(T)$

2. T 1-1 $\Leftrightarrow \text{nullity}(T)=\{0\} \Leftrightarrow$ onto (if $\dim(V)=\dim(W)$)

$T: V\to W$

3. $x\in V$, $\beta=\{v_1\cdots v_n\}$, $x=\sum a_i v_i$

$[x]_\beta=\begin{pmatrix}a_1\\ \vdots\\ a_n\end{pmatrix}$ x 在基 β 下的坐标向量.

$[T]^\gamma_\beta[x]_\beta=[T(x)]_\gamma$ → ex. assignment. L.T:

↓

特例. $[T]_\beta[x]_\beta=[T(x)]_\beta$

$A=\begin{pmatrix}| & & |\\ [T(v_1)]_\gamma & \cdots & [T(v_n)]_\gamma\\ | & & |\end{pmatrix}$

4. $(AB)C=A(BC)$

矩阵乘法结合律

(顺序不可换、括号可加).

5. $tr(AB)=tr(BA)$

proof: 设 $A_{m\times n}$, $B_{n\times m}$

$(AB)_{ii}=\sum_{j=1}^{n}A_{ij}B_{ji}$

$tr(AB)=\sum_{i=1}^{m}(AB)_{ii}=\sum_{i=1}^{m}\sum_{j=1}^{n}A_{ij}B_{ji}$

$\begin{pmatrix}a_{11} & a_{1m}\\ a_{n1} & a_{nn}\end{pmatrix}$ $(BA)_{jj}=\sum_{i=1}^{m}B_{ji}A_{ij}$

$tr(BA)=\sum_{j=1}^{n}(BA)_{jj}=\sum_{j=1}^{n}\sum_{i=1}^{m}B_{ji}A_{ij}$

6. 1-1, onto, finite-dimension $\Leftrightarrow T^{-1}$ exist.

7. $([T]^\gamma_\beta)^{-1}=[T^{-1}]^\beta_\gamma$

变换矩阵的逆=逆变换的矩阵

8. $tr(P^{-1}AP)=tr(A)$ 相似矩阵迹相等

proof: $tr(P^{-1}(AP))=tr(A)=tr((AP)P^{-1})$

↑

$tr(XY)=tr(YX)$.

$B=P^{-1}AP$

线变: ↓ ↓ ↓ ↓

$[T]^\beta_\beta=[I]^\beta_\alpha[T]^\alpha_\alpha[I]^\alpha_\beta$ 从右往左读

9. if $AB=I_n$, $A,B\in M_{n\times n}$

show that $A=B^{-1}$.

$AB=I_n$

$n=\text{rank}(I_n)=\text{rank}(AB)\le\text{rank}(B)\le n$

so $\text{rank}(B)=n$. B invertible.

$AB=I_n$

$ABB^{-1}=I_nB^{-1}$, $AI_n=B^{-1}$, $A=B^{-1}$

1. $\text{rank}(M)$ 代表矩阵 M 的秩, 矩阵线性无关列/行向量最大个数

2. AB 先做 B 后做 A, $\text{Im}(AB)\subseteq\text{Im}(A)$

$(AB)x=A\cdot(Bx)$ so $A\,\text{Im}(B)=\text{Im}(AB)$

apply A will eliminate some entries,

so $\dim\text{Im}(AB)\le\dim\text{Im}(B)$.

$\text{rank}(AB)\le\min\{\text{rank}(A),\text{rank}(B)\}$

向量可以个或少, 但维数不会增加, 因为维数

↓ 基个数

↓ 线性无关向量的个数 max

线性无关向量可以一直到到 $\mathbb{R}^n$ 方向. 但不会

↑

10. 对于方阵, 满秩 $\Leftrightarrow$ 可逆.

proof: left as an exercise.

满秩$\Rightarrow$可逆: n 个 L.I.D 可构成 Basis

对 I_n 的每一列 e_i, 一定可找到 c_i

使 $Ac_i=e_i$

↓

拼到拼好 C. $AC=I_n$, $CA=I_n$

A invertible.

$\text{rank}(A)=\dim\text{Im}(A)=n$

$\text{Im}(A)\subseteq\mathbb{R}^n$

$\text{Im}(A)=\mathbb{R}^n$

$\text{Im}(A)=\{Ax\mid x\in\mathbb{R}^n\}=\mathbb{R}^n$, $b\in\mathbb{R}^n=\text{Im}(A)$, $Ax=b$

Replacement Thm. + finite dimension.

1. $\dim V=n$, $S\subseteq V$

推论：

1° $\text{span}(S)=V$, $|S|\geq n$.

2° S linearly independent, $|S|\leq n$.

i) $|S|=n$

ii) $\text{span}(S)=V$

iii) S linearly independent

} 2 => 1 -> 就是一组basis

知二求三

2. Extend Thm. (W的一组基，一定可以往里面继续加向量扩充成整个V的一组基)

$W\subseteq V$, $\dim(W)\leq\dim(V)$

而且 if $W\subseteq V$ + $\dim(W)=\dim(V) \Rightarrow W=V$

子空间维数=母空间，共轭无它自己.

prove: $\dim(W_1+W_2)=\dim(W_1)+\dim(W_2)-\dim(W_1\cap W_2)$

idea: 证 dimension (Def. basis 的 number).

$W_1\cap W_2$ -> basis $\{v_1\cdots v_k\}$

n m

↓

Extend $\{v_1\cdots v_k u_{k+1}\cdots u_n\}$ W_1

$\{v_1\cdots v_k r_{k+1}\cdots r_m\}$ W_2.

↓

$\beta=\{v_1\cdots v_k u_{k+1}\cdots u_n r_{k+1}\cdots r_m\}$

suppose $\sum a_i v_i+\sum b_i u_i+\sum c_i\gamma_i=0$

一个向量 $=\sum a_i v_i+\sum b_i u_i=-\sum c_i r_i\in W_1\cap W_2$

同时$\in W_1$ 和 W_2 ↓$\in W_1$ ↓$\in W_2$

$-\sum c_i\gamma_i=\sum d_j v_i$

$\sum d_i v_i+\sum c_i\gamma_i=0$. -> basis of W_2.

all $c_i, d_i=0$.

substitude back $\sum a_i v_i+\sum b_i u_i=0$.

all $a_i, b_i=0$.

so β is linearly independent.

$w\in W_1+W_2$.

$w=w_1+w_2$.

$\text{Span}(\beta)\geq W_1+W_2$

$\subseteq$

$\text{Span}(\beta)=W_1+W_2$

Thus β is basis of W_1+W_2

The proof is over.

Concept in Matrix:

1. symmetic matrix $A^T=A$, A is square matrix, $A_{ij}=A_{ji}$
2. diagonal matrix $\begin{pmatrix} \backslash & 0 \\ 0 & \backslash \end{pmatrix}$
3. upper triangular matrix $\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$ 主对角线下方全0.
4. trace: $\sum a_{ii}$ (方阵)
5. Identity matrix $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
6. rank $\text{rank}(A)=\dim\text{Row}(A)=\dim\text{Col}(A)=\text{rank}(A^T)$
7. determinant (只定义在方阵上，输出一个数) 方阵可逆 $\Leftrightarrow \det(A)\neq 0$
8. ~~invertible~~ / inverse A^{-1} Equivalent $\Leftrightarrow$ 满秩，$\det\neq 0$，column vectors L.I.D, one-to-one + onto
9. elementary row/column operation,

$AB=I_n \Rightarrow BA=I_n$

prove:

$(BA)B=BI_n=B=I_nB$

$BA=I_n$.

$(BA-I_n)B=0$. 只有零变换能把一组线性无关的全映成0,

D: 到底示. 到无关

$\text{rank}(I_n)=n=\text{rank}(AB)\leq\text{rank}(B)\leq n$

$\text{rank}(B)=n$.

($\text{Ker}(B)=\{0\}$, $Bx=0$ has only trivial solution) or

10. elementary matrix.
(I经过一次 elementary operation)
11. argumented matrix $(A|B)$
12. $Ax=b$ 非齐次线性方程组：解决能把b表成A的列的线性性组合的系数有谁

$Ax=0$ 齐次方程组：解 = ker(A)，永远至少有0解

Relation: 非齐次一个特解 + 齐次全部解 = 非齐次通解

$Ax_p=b$, $Ax_h=0$, $x_h\in V$

$Ax=b$, $x=x_p+x_h$

$A(x_p+x_h)=b+0=b$.

反过来，$Ax_1=b$, $Ax_2=b$

$A(x_1-x_2)=b-b=0$.

$x_1-x_2=x_h$

$T(v)=w$ for $\forall v$.

$\text{Im}(T)$

$=\{T(v)|v\in V\}$

$\text{Im}(T)=W$.

one-to-one + onto

$\ker(T)=\{0\}$, $\text{Im}(T)=W$

$T: V\to W$

$T(v_1)=T(v_2)\to T(v_1)-T(v_2)=0$

$v_1=v_2$ $\quad T(v_1-v_2)=0$

$v_1-v_2=0$.

so $T(0)=0$

*8. $T: V\to W$, $\dim V=\dim W$

1-1 $\Leftrightarrow$ on-to $\Leftrightarrow$ 可逆

Apply to matrix: 方阵.

列无关(1-1) $\Leftrightarrow$ 列张成全空间 $\Leftrightarrow$ invertible

(onto)

13. RREF
1. all zero row below
2. pivot=1 和row 且在上一行pivot右侧
3. 在一行内，除了pivot=1. 其它全0.

REF. 无第3条.

Matrix:

1. row_op <=> left multiply $A \Rightarrow EA$

column-op <=> right multiply. $A \Rightarrow FA$

2. $\text{rank}(AB) \le \text{rank}(A), \text{rank}(B)$.

$\text{rank}(UT) \le \text{rank}(U), \text{rank}(T)$.

$\text{rank}(PAQ) = \text{rank}(A)$. P, Q are invertible.

proof: $\text{rank}(PAQ) \le \text{rank}(AQ) \le \text{rank}(A)$.

$\text{rank}(A) = \text{rank}(P^{-1}PAQQ^{-1}) \le \text{rank}(PAQQ^{-1})$

$\le \text{rank}(PAQ)$.

3. $(A|I_n) \to (I_n|A^{-1})$

4. $\text{rank}(A) = \text{rank}(A^T)$ ☆

转置矩阵秩不变.

行秩=列秩.

5. $Ax=b$ has solution <=> $b \in \text{col}(A)$.

$m\times n$ $n\times 1$ $m\times 1$

b为A列空间的 <=> b可写成A的列向量的线性组合.

$A = \begin{pmatrix} | & | & & | \\ a_1 & a_2 & \cdots & a_n \\ | & | & & | \end{pmatrix}_{m\times n} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n\times 1} = \underbrace{x_1a_1 + x_2a_2 + \cdots + x_na_n}_{m\times 1} = b$

每一个 a_i 本质是 $m\times 1$ 列向量

6. RREF unique.

7. $\text{rank}(T) = \text{rank}([T]_B^\gamma)$

上下列秩和选定哪组基会出现代表矩阵元素.

8. 矩阵乘法3种理解方式. (Product = 3 kinds of understanding)

1. $\begin{pmatrix} | & | & & | \\ a_1 & a_2 & \cdots & a_n \\ | & | & & | \end{pmatrix}_A \begin{pmatrix} & & i & \\ & & \vdots & \\ & & b_i & \\ & & \vdots & \end{pmatrix}_B = \begin{pmatrix} & & i & \\ & & \vdots & \\ & & AB & \\ & & \vdots & \end{pmatrix}$

AB的第i列就是:

用B的第i列里的数字当作系数,对A全部列向量线性组合拼出来.

2. $\begin{pmatrix} - & \\ & \\ & \end{pmatrix}_A \begin{pmatrix} | & & | \\ b_1 & \cdots & b_n \\ | & & | \end{pmatrix}_B = \begin{pmatrix} | & & | \\ AB & \cdots & AB \\ | & & | \end{pmatrix}$ i

AB第i行:

用A的第i行作系数,对B all 行向量线性组合拼出来

Linear Transformation:

1. $(T+U)(x) = T(x) + U(x)$.

2. $(aT)(x) = aT(x)$.

3. $UT = U \circ T = (UT)(x) = U(T(x))$

4. T^{-1}, invertible.

$T: V \to W$, $U: W \to V$.

满足both: 1. $TU = I_W$ 为W上的恒等

2. $UT = I_V$ 为V上的恒等.

3. similar matrices (det, tr, rank 均等)

$B = P^{-1}AP$ 找出一个 P 满足即可

Determinants:

$$\det(A) = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} \det(\widetilde{A_{ij}})$$

↑ 任选一行，一列展开.

↓ 余子式：删去第 i 行，第 j 列

ex. 按第一列展开.

$$\det(A) = \sum_{i=1}^{n} (-1)^{1+i} a_{i1} \cdot \det(\widetilde{A_{i1}})$$

$$A = \begin{pmatrix} 2 & 1 & 4 \\ 3 & 5 & 6 \\ 1 & 0 & 7 \end{pmatrix}$$

$$\det(A) = a_{11}(-1)^{1+1} \det\begin{pmatrix} 5 & 6 \\ 0 & 7 \end{pmatrix} + a_{12}(-1)^{1+2} \det\begin{pmatrix} 1 & 4 \\ 0 & 7 \end{pmatrix} + a_{13}(-1)^{1+3} \det\begin{pmatrix} 1 & 4 \\ 5 & 6 \end{pmatrix}$$

$$= 2(35-0) - 3(7-0) + 1(6-20)$$

$$= 70 - 21 - 14 = 35$$

$$\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

推理:

1. $\det(A) \neq 0 \iff A$ invertible, 满秩

2. $\det(AB) = \det(A) \cdot \det(B)$.

3. $\det(P^{-1}AP) = \det(A)$

1. row op. doesn't change column's linear relationship.

2. 性质: 1° row/column linear

$$\det\begin{pmatrix} | & | & | \\ \alpha_1 & au+bv & \alpha_3 \\ | & | & | \end{pmatrix} = a\det\begin{pmatrix} | & | & | \\ \alpha_1 & u & \alpha_3 \\ | & | & | \end{pmatrix} + b\det\begin{pmatrix} | & | & | \\ \alpha_1 & v & \alpha_3 \\ | & | & | \end{pmatrix}$$

对单独一列/行，其它列/行不变

ex. $\det\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} + \det\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} = \det\begin{pmatrix} 1 & 3 & 0 \\ 0 & 2 & 1 \\ 1 & 2 & 1 \end{pmatrix}$

4. $\det(A) \neq 0$

$\iff \operatorname{rank}(A) = n$

$\iff A$ is invertible.

2° row extend /column

ex. $k \cdot \det\begin{pmatrix} \square \end{pmatrix} = \det\begin{pmatrix} k \\ \vdots \\ k \end{pmatrix}$

$\det(\cdots, kr_i, \cdots) = k \cdot \det(\cdots, r_i, \cdots)$.

3° i) row/column identical 两列/行完全一样.

$\det(A) = 0$

ii) exchange 2 rows/columns

$\det(A) = -\det(B)$.

iii) adding a multiple of one row to another.

一行倍数加在另一行上(列)

$\det(A) = \det(B)$

5. (upper) triangular matrix.

$$\det = \prod_{i=1}^{n} a_{ii}$$

compute: 故求 det. 又用 iii) 不改变 det.

trick

↑ ex. $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 0 & 0 \\ 2 & 1 & 0 \end{pmatrix}$

↓ $\begin{pmatrix} 1 & 2 & 3 \\ 0 & -8 & -12 \\ 0 & -3 & -6 \end{pmatrix} \to \begin{pmatrix} 1 & 2 & 3 \\ 0 & -8 & -12 \\ 0 & 0 & -\frac{3}{2} \end{pmatrix}$

$\det(A) = 8 \times \frac{3}{2} = 12$

(不要交换行，也不要单行乘倍)